Chapter test

34 Regulation of gas content in blood

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following comparisons between voluntary and involuntary control of breathing is ***incorrect***?

	voluntary control	***involuntary control***
A.	for maintaining a steady breathing rhythm	for regulating gas content in blood
B.	coordinated by the cerebrum	coordinated by the medulla oblongata
C.	can stop breathing	cannot stop breathing
D.	conscious	unconscious

2. Which of the following statements about the respiratory centre is/are ***incorrect***?

(1) The respiratory centre is located in the cerebral cortex.
(2) The respiratory centre receives sensory information from chemoreceptors.
(3) The respiratory centre passes motor information to the respiratory muscles.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

3. Which of the following statements about the initiation of heartbeat are ***incorrect***?

(1) The initiation of heartbeat is myogenic.
(2) Both the SA node and the AV node generate electrical impulses to initiate heartbeat.
(3) The initiation of heartbeat is unaffected by the nervous system.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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4. Which of the following comparisons between SA node and AV node is ***incorrect***?

	SA node	***AV node***
A.	located in the wall of the right atrium	located at the base of the right atrium
B.	connected with the bundle of His	connected with the sympathetic nerve
C.	generates electrical impulses	relays electrical impulses
D.	impulses from it cause the two atria to contract	impulses from it cause the two ventricles to contract

5. Which of the following comparisons between atrial systole and ventricular systole is correct?

	Atrial systole	***Ventricular systole***
A.	lasts longer in time	lasts shorter in time
B.	bicuspid and tricuspid valves are closed	bicuspid and tricuspid valves are open
C.	heart sound is produced	no heart sound is produced
D.	blood flows from atria to ventricles	blood flows from ventricles to arteries

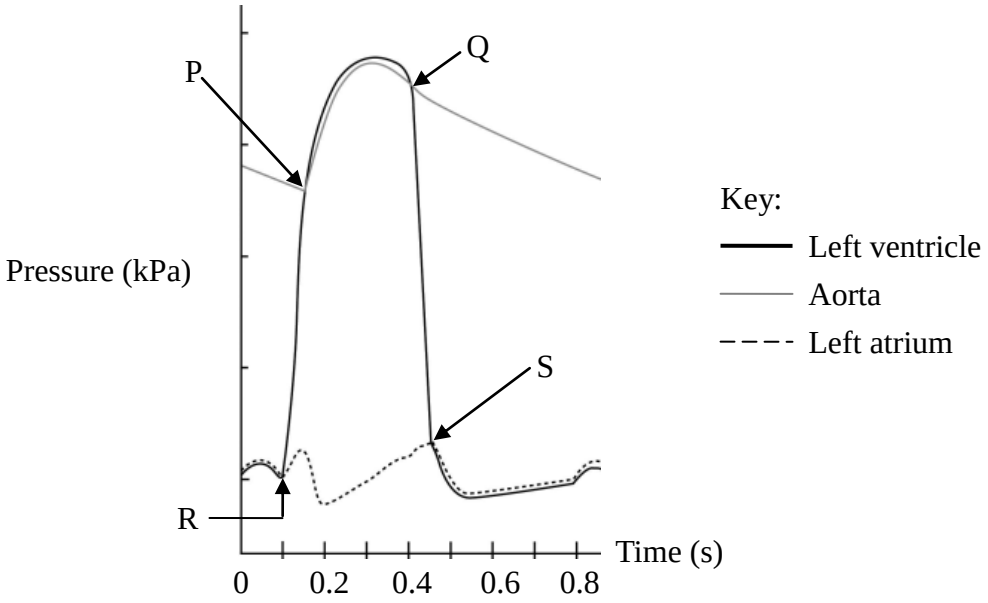
6. Which of the following statements about diastole in the cardiac cycle are correct?

(1) Diastole is the longest phase in the cardiac cycle.
(2) Heart sound is produced during diastole.
(3) Blood flows from atria to ventricles during diastole.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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★ 7. The graph below shows the pressure changes in the left ventricle, aorta and left atrium during a cardiac cycle.



What are the conditions of the semilunar valves and atrioventricular valves at P, Q, R and S?

	Semilunar valves		***Atrioventricular valves***	
	Open	***Close***	***Open***	***Close***
A.	P	Q	R	S
B.	P	Q	S	R
C.	Q	R	P	S
D.	Q	R	S	P

8. Which of the following statements about hormonal control of the cardiac output are correct?

(1) Adrenaline is a hormone which stimulates the SA node and cardiac muscle.
(2) Adrenal gland increases its secretion of adrenaline upon stimulation from the sympathetic nerve.
(3) Adrenaline increases the cardiac output by increasing the heart rate and stroke volume.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

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★ 9. Which of the following contribute to the increase in total cardiac output during exercise?

(1) vasodilation of blood vessels in the skin
(2) increase in stroke volume
(3) increase in heart rate

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

10. During exercise, which of the following changes will occur?

(1) The respiratory centre is stimulated to increase the depth of breathing.
(2) The cardiovascular centre is stimulated to reduce the cardiac output.
(3) Adrenal glands are stimulated to secrete more adrenaline.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	34.2	34.2	34.3	34.3	34.3	34.3	34.3	34.3	34.4	34.4

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B. Problem solving (18 marks)

★ 11. In an investigation to study the effect of oxygen concentration and carbon dioxide concentration on the rate of breathing, a healthy student was asked to breathe in gas mixtures of different compositions. His breathing rate under different conditions was measured and recorded in the following table.

Gas mixture	Oxygen concentration	Carbon dioxide concentration	Rate of breathing (breaths per minute)
I	21%	0.04%	18
II	16%	0.04%	18
III	21%	4%	36
IV	16%	4%	36

(a) With reference to the results, deduce whether oxygen concentration or carbon dioxide concentration is the major factor that affects the rate of breathing. (3 marks)

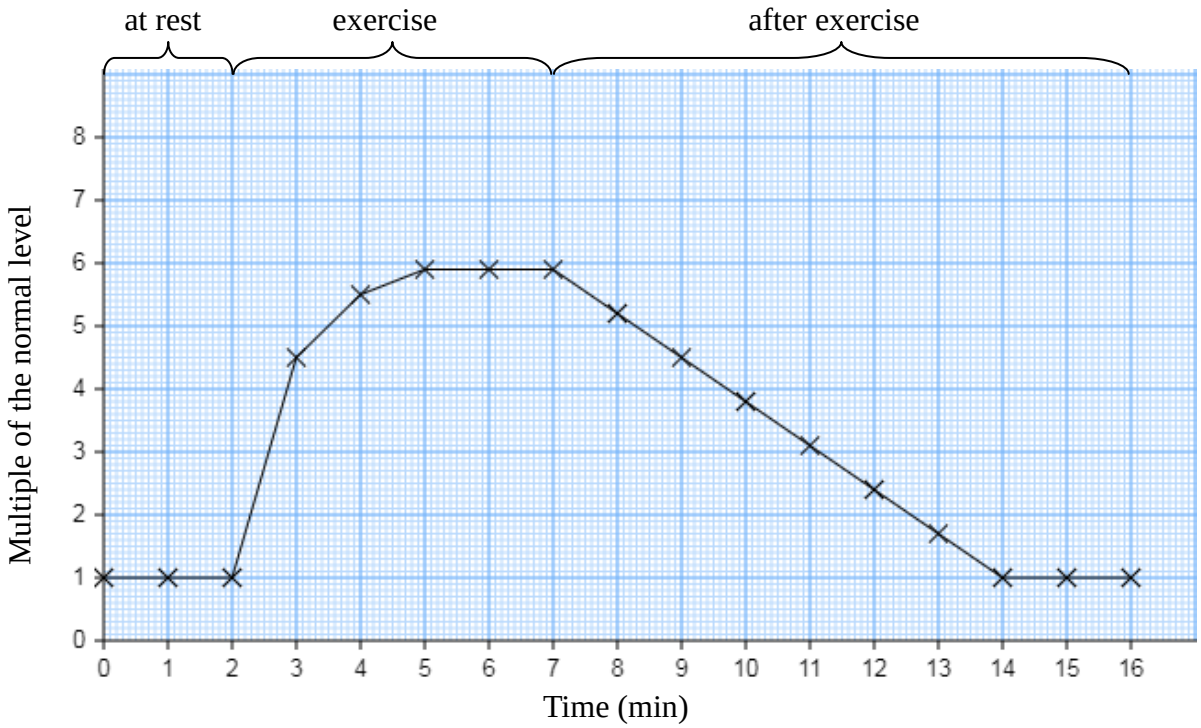
(b) Which gas mixture has a composition similar to exhaled air? (1 mark)

(c) Mouth-to mouth ventilation can be performed to restore ventilation in a person who has stopped breathing. Explain the physiological basis of this procedure. (4 marks)

(Total: 8 marks)

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★ 12. The graph below shows the changes in the rate of oxygen uptake by a person at rest, during exercise and after exercise.



(a) The time taken after exercise for the rate of oxygen uptake to return to the normal level is called the recovery time. Determine the recovery time of the person from the graph. (1 mark)

(b) (i) Apart from the rate of oxygen uptake, the blood lactic acid concentration of the person also changes during exercise. What process leads to the production of lactic acid in the body during exercise? (1 mark)

(ii) Sketch a curve in the graph to show how the blood lactic acid concentration changes during and after exercise. (1 mark)

(iii) Explain the pattern you sketched in (ii). (4 marks)

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(c) Based on the graph and your biological knowledge, describe how the body removes the lactic acid accumulated in the body. (3 marks)

(Total: 10 marks)

C. Exam challenge (12 marks)

★★ 13. In an investigation to study the effect of exercise on the heart rate and stroke volume, two students were asked to carry out a series of step-up exercises with increasing intensity. Their heart rate and stroke volume were measured immediately after each set of exercise. The students were allowed to rest for 15 minutes before doing exercise of the next intensity level. The results are shown below.

Student	Heart rate after exercise (beats per minute)			
	Resting	2 step-ups in 10s	4 step-ups in 10s	8 step-ups in 10s
A	64	65	96	134
B	73	81	129	152

Table 1

Student	Stroke volume after exercise (cm^3)			
	Resting	2 step-ups in 10s	4 step-ups in 10s	8 step-ups in 10s
A	98	100	125	140
B	70	83	92	118

Table 2

(a) Calculate the cardiac output of student A after doing 8 step-ups in 10s. Show your working. (2 marks)

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(b) With reference to the results, describe the effect of increasing intensity of exercise on the heart rate and the physiological regulation of this response. (5 marks)

(c) State the significance of the change described in (b). (3 marks)

(d) Suggest which student is in the sports team and practices running every day. Give a reason to support your answer. (2 marks)

(Total: 12 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	A	2.	A	3.	C	4.	B	5.	D
6.	D	7.	B	8.	D	9.	C	10.	B

(1 mark each)

B. Problem solving (18 marks)

11. (a) Carbon dioxide concentration (1)
The rate of breathing increased when the carbon dioxide concentration in the gas mixture was increased. (1)
The rate of breathing remained unchanged when the oxygen concentration in the gas mixture was varied. (1)

(b) Gas mixture IV (1)

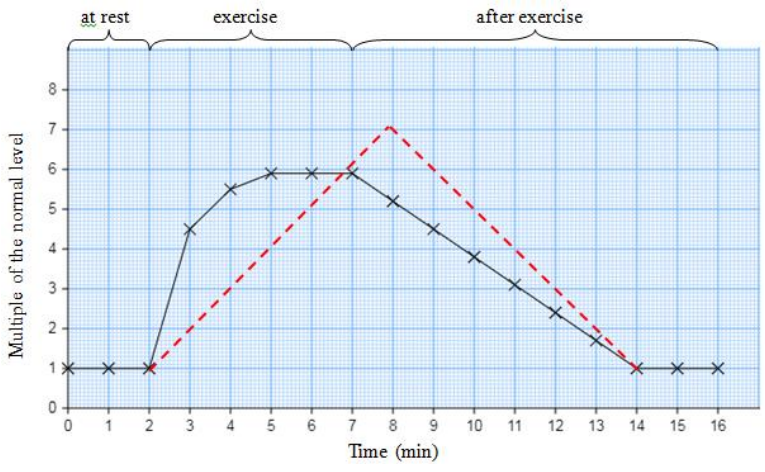
(c) When exhaled air (containing a higher carbon dioxide concentration) is blown into the person's lungs, more carbon dioxide enters the person's bloodstream. (1)
This increased carbon dioxide concentration is detected by chemoreceptors (1)
which then send nerve impulses to stimulate the respiratory centre in the medulla oblongata. (1)
The respiratory centre sends more nerve impulses (via a sympathetic nerve) to the intercostal muscles and diaphragm muscles. (1)
The increased nervous stimulation to these respiratory muscles could help restore breathing.

(Total: 8 marks)

12. (a) 7 minutes (1)

(b) (i) anaerobic respiration (1)

(ii) (1)



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(iii) When the oxygen supply cannot meet the demand for aerobic respiration, (1)
muscle cells undergo anaerobic respiration to release extra energy in order to cope with immediate energy need. (1)
During exercise, the rate of lactic acid production is faster than the rate of lactic acid breakdown, so the level of lactic acid in the blood increases. (1)
After exercise, the rate of lactic acid breakdown is faster than the rate of lactic acid production, so the level of lactic acid in the blood decreases. (1)

(c) After exercise, the rate of oxygen uptake remains higher than normal for some time. (1)
This enables the body to take in more oxygen (1)
to oxidize / break down lactic acid to carbon dioxide and water (or convert lactic acid to glycogen). (1)

(Total: 10 marks)

C. Exam challenge (12 marks)

13. (a) 134×140 (1)
$= 18760\ cm^3$ per minute (1)

(b) The heart rate increases with increasing intensity of exercise. (1)
The rate of respiration of muscle cells increases as the intensity of exercise increases.
More carbon dioxide is produced and the blood pH decreases. (1)
The cardiovascular centre is stimulated. (1)
It sends more nerve impulses to the SA node of the heart via the sympathetic nerve. (1)
The cardiac muscles contract more frequently, thus the heart rate increases. (1)

(c) The increase in the heart rate increases the rate of blood flow. (1)
This helps transport more oxygen to the muscle cells for aerobic respiration. (1)
More carbon dioxide is transported to the lungs for faster removal. (1)

(d) Student A (1)
The stroke volume of student A is higher than that of student B. Regular training strengthens the cardiac muscle, so a greater pumping force is produced when the cardiac muscle contracts, more blood is pumped out per heartbeat. (1)

(Total: 12 marks)

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